

Malik Hashim

Senior Software Engineer — Full-Stack & AI | Web, Mobile & Cloud

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Summary

Senior software engineer with 5+ years building and shipping **full-stack products end-to-end** — backends (NestJS, Node, PostgreSQL, Prisma), web frontends (React, Next.js), cross-platform mobile (React Native, Flutter), and the AWS / CI-CD pipelines that deliver them. I add **LLM-powered features** — chat, retrieval over user data, voice, and agentic workflows — from model integration to production. 10+ products shipped across AI-powered health apps, real-time surveillance and ride-sharing platforms, and enterprise white-label products, with team leadership and direct client ownership.

Work Experience

Lead Software Engineer | Technical Lead

2024 – Present

Verasana Inc., United States, Remote

- Lead a team of 3 with full ownership of **8 products end-to-end** for US fitness and wellness businesses — technical architecture, full-stack delivery, and release.
- Build cross-platform clients in **React Native and Flutter**, backed by integrated production services — **Mux** (video), **Stream** (chat), Google Cloud Storage (media), and Sentry (monitoring).
- Ship production features per client — **performance dashboards, video courses, in-app chat, and FastTrack programs**.
- Own the full release pipeline — automated iOS/Android deployments via **Fastlane**, App Store submissions, and CI/CD.
- Manage 1-on-1 client relationships, aligning scope, timelines, and deliverables.

Tech Stack:

React Native, Flutter, TypeScript, Dart, Firebase • Mux, Sentry, Google Cloud Storage, Stream, Fastlane, Xcode, CI/CD

Senior Full-Stack Engineer

2022 – 2023

CRV Surveillance, United States, Remote

- Built the complete product from scratch — **backend (NestJS, PostgreSQL, Prisma), admin web portal (React / Next.js)**, and Flutter mobile app.
- Integrated **live CCTV feed streaming** into the mobile app and web portal via WaveSync Server with low-latency optimization.
- Developed real-time **interactive maps (Google Maps API)** for tracking technician and contractor field locations.
- Built modules for **technician/contractor management, job assignments, and site visits** with full field activity reporting.
- Developed a full **invoicing and billing system** — invoice generation, payment records, status tracking, and contractor payouts.
- Implemented **MSAL authentication** with role-based access control for admins, technicians, and contractors.

Tech Stack:

NestJS, React, Next.js, Flutter, PostgreSQL, PrismaORM • WaveSync, Google Maps API, MSAL Auth, AWS, Docker, CI/CD

Full-Stack Engineer

2022

Atualizado, Remote

- Built a **ride-sharing platform end-to-end** — MERN / Node backend with PrismaORM and PostgreSQL, plus a cross-platform Flutter client — with real-time tracking, dynamic routes, and bus-route merging.
- Implemented **Dijkstra's shortest-path algorithm** for bus routing, achieving 30% faster travel times.
- Used **WebSockets (60% faster updates)**, Redis, AWS CloudFront, and CI/CD pipelines for scalable deployment.
- Integrated Google Maps API for fare estimation, geocoding, and distance management; Firebase for push notifications.
- Built Driver and Passenger modules covering ride requests, trips, and live location updates.

Tech Stack:

Node.js, MERN Stack, PrismaORM, PostgreSQL, Redis, Flutter • WebSockets, Google Maps API, Firebase, AWS CloudFront, CI/CD

Web & Mobile Engineer

2021

HHRC Digital Dubai EDS, Court Cases Management Department, Remote

- Built a **Flutter companion app** and contributed to the **Liferay + React web platform** for HHRC's court case management system.
- Created **microfrontend packages** for modularity and easy cross-team integration on the web platform.
- Implemented **Dynamic Form Builders** with custom logic for seamless case creation, editing, and review workflows.
- Managed Jira ticketing with story points to keep delivery on track.

Tech Stack:

React, Liferay, Flutter, Microfrontend Architecture, Jira

Software Engineer — Conversational AI

2021

AutoVoice AI, Remote

- Built **AI-driven conversational chatbot workflows (BotPress)** powering automated, in-app user engagement.
- Integrated **Twilio A2P** messaging for seamless in-app communication.

Tech Stack:

BotPress, Twilio A2P, Flutter, Firebase

Full-Stack Developer

2020

Nook, NUST Islamabad

- Built the frontend in **React Native** and migrated the backend from Python (Django/Flask) to **NestJS**.
- Designed the database schema and overall system architecture.
- Implemented **CI/CD pipelines** with AWS CodeBuild, CodePipeline, and Elastic Beanstalk for staging and production.

Tech Stack:

NestJS, React Native, PostgreSQL, AWS CodeBuild, CodePipeline, Elastic Beanstalk

Software Engineer

2019

NephroFit (Chronic Kidney Disease App), NUST Islamabad

- Led a team of 3 to ship a Flutter + Firebase CKD management app featuring **ChatGPT / LLM API integration** for patient guidance and **ML-based health prediction and monitoring** (Inception V3, Bayesian networks).
- Turned model outputs into real-time, user-facing health insights, measurably improving patient engagement.

Tech Stack:

Python, Flutter, Firebase, ChatGPT API, Machine Learning

Skills

AI / LLM Engineering OpenAI & Anthropic APIs • Streaming & token-by-token UX • Function calling / tool use • Agentic workflows • RAG & vector search (pgvector, Pinecone) • Prompt engineering • Voice & speech-to-text (Whisper / realtime APIs)

Languages TypeScript • JavaScript • Dart • Python • SQL

Backend & APIs NestJS • Node.js • Python • PostgreSQL • PrismaORM • pgvector • Firebase • REST APIs • WebSockets • Redis

Web & Frontend React • Next.js • TypeScript • JavaScript • HTML/CSS • MERN Stack • Microfrontend Architecture

Mobile React Native • Flutter • Dart • iOS (Xcode) • Android • Fastlane • App Store / Play Store

Cloud & DevOps AWS (EC2, S3, CloudFront, CodeBuild, CodePipeline) • Google Cloud • Docker • CI/CD

Integrations & Tools Google Maps API • Mux • Sentry • Twilio • Stream • BotPress • MSAL Auth • Liferay • Jira • Agile / Scrum

Foundations Data Structures & Algorithms • System Design • Machine Learning (OpenCV, NumPy, Scikit-image, Inception V3, Bayesian Networks)

Certifications & Education

Bachelor of Computer Science — NUST Islamabad

2018 – 2022

- **Career Essentials in Generative AI** — Microsoft & LinkedIn
- Bayesian Networks — **NPU** Global Open Courses Program
- Care Foundation — **Marketing & Fundraising** • **SAKURA Science Exchange Program** — JST (Japan Science & Technology)